Lesson 1: Variables, print() and input()

The complete lesson - concept, full program, walkthrough, verified outputs, common mistakes, exercises with solutions, and knowledge check. From the Python-for-Pharmacy-Students repo (01-variables), plus notes from the live session on 26 Oct 2026. Open in Samsung Notes to read and annotate.

1. The concept in plain words

A variable is a labelled box that holds a value. weight_kg = 60 means: 'remember 60 under the name weight_kg'.

print() shows something on the screen. input() asks the person running the program to type something in.

One catch: input() always gives back TEXT, even if the person typed a number. So float(input(...)) converts that text into a number you can do maths with.

Prerequisites: none. This is the first lesson - everything starts here.

2. The pharmacy analogy (the dabba idea)

Think of a steel dabba (box) in the kitchen. You stick a label on the outside and put something inside. That is a variable. drug = "Paracetamol" sticks the label 'drug' on a box and puts the word Paracetamol inside. print(drug) - with no quotes - opens that box and shows what is inside.

The = sign does NOT mean 'equals' like in maths. It means 'store the right side inside the left side'.

Quotes rule: quotes = the thing itself (text). No quotes = go find the box with that name. That is why print(Amoxicillin) with no quotes gives a NameError: Python looks for a box called Amoxicillin and finds none.

3. The pharmacy task

A paracetamol dose calculator. Paracetamol is commonly dosed at 15 mg per kg of body weight per dose, capped at a maximum single dose of 1000 mg. The script asks for a weight, works out the dose, and tells you how many 500 mg tablets that is.

Why the dosing rule lives in variables: if the drug changes (for example to ibuprofen: 10 mg/kg, max 400 mg), you change two numbers in one place instead of hunting through the whole program. That is the real reason programmers use variables for rules.

4. The full program (repo file: 01-variables/dose_calculator.py)

```
# Lesson 1: Variables, print() and input()
```

```
# Ask the user for the patient's weight.
# input() always gives back TEXT, so float() converts it to a number.
weight_kg = float(input("Enter patient weight in kg: "))

# Store the dosing rule in variables, so the numbers are easy to change.
mg_per_kg = 15        # paracetamol dose per kg of body weight
max_dose_mg = 1000    # maximum single dose, in mg

# Calculate the dose. min() picks the smaller of the two values,
# so the dose never goes above the maximum.
dose_mg = min(weight_kg * mg_per_kg, max_dose_mg)

# A 500 mg tablet is the common tablet strength.
tablet_strength_mg = 500
tablets = dose_mg / tablet_strength_mg

# Show the result.
print("Patient weight:", weight_kg, "kg")
print("Paracetamol dose:", dose_mg, "mg")
print("That is", tablets, "tablet(s) of 500 mg")
```

How to run it: python3 dose_calculator.py (or the Run button in VS Code). Type a weight when it asks, and press Enter.

5. Line-by-line walkthrough

- weight_kg = float(input(...)) - show the question, wait for typing, convert the typed text into a number, store it in the box 'weight_kg'.

- mg_per_kg = 15 - store the dosing rule (15 mg per kg) in a box, so it is easy to change later.

- max_dose_mg = 1000 - store the safety maximum (1000 mg) in a box.

- dose_mg = min(weight_kg * mg_per_kg, max_dose_mg) - multiply weight by 15, then min() keeps the smaller of the result and 1000. The dose can never cross the maximum. Store the final dose in 'dose_mg'.

- tablet_strength_mg = 500 - store the tablet strength.

- tablets = dose_mg / tablet_strength_mg - divide the dose by 500 to get the number of tablets.

- The three print lines - show the weight, the dose, and the tablet count. print() can show several things separated by commas, and it puts a space between them.

6. Expected vs verified output

Type 60:

```
  Enter patient weight in kg: 60
```

```
Output: Patient weight: 60.0 kg / Paracetamol dose: 900.0 mg / That is 1.8 tablet(s) of 500 mg.
```

Type 70:

```
  Enter patient weight in kg: 70
```

```
Output: Patient weight: 70.0 kg / Paracetamol dose: 1000.0 mg / That is 2.0 tablet(s) of 500 mg.
```

Why 70 gives 1000 and not 1050: 70 x 15 = 1050, but min(1050, 1000) keeps 1000. The safety ceiling works. Both outputs above were verified by running the program.

7. Pharmacy application

This is not a toy example. Weight-based dosing with a maximum cap is exactly how paracetamol is prescribed, and the program structure is exactly how real clinical calculators work: take an input, apply a rule stored in variables, enforce a safety limit, and report a usable answer (tablets, not just milligrams). Changing the drug means changing the rule variables - the rest of the program stays the same.

8. Common mistakes (all seen in the live session)

- NameError - using a name with no box behind it, e.g. print(Amoxicillin) with no quotes. Fix: quotes for text, or create the box first.

- ValueError - asking float() to convert something that is not a number, e.g. a file path fed in as the 'answer'. Fix: type only a number when it asks.

- KeyboardInterrupt - not a code error. Ctrl+C pressed while the program was waiting at input(). The program was fine; it was stopped from outside.

- Answer inside the question - writing the number inside the input message, e.g. input("Enter weight: 70"). The message is only the question; the answer is typed AFTER the program runs and asks.

- Assignment inside print() - print(dose = 1000) is wrong. print() only shows values; it cannot store them. Store first (dose = 1000), then print(dose).

9. Practice exercises (try first, then peek at the solutions)

Exercise 1 (pharma): Change the dose rule to ibuprofen: 10 mg/kg per dose, maximum 400 mg. Run it for a 70 kg patient. What dose do you get?

Hint: only two numbers change - mg_per_kg and max_dose_mg.

Exercise 2 (intermediate): Add a line that also prints the dose in grams (1 g = 1000 mg).

Hint: divide the mg dose by 1000 and store it in a new box before printing.

Exercise 3 (beginner + pharma): Ask for the patient's name too, and print it in the result, for example: 'Dose for Rahul: 900 mg'.

Hint: a name is text, so it needs input() WITHOUT float(). Then print it with the dose.

10. Solutions (only after trying)

Solution 1 - ibuprofen dosing:

```
weight_kg = float(input("Enter patient weight in kg: "))
mg_per_kg = 10
max_dose_mg = 400
dose_mg = min(weight_kg * mg_per_kg, max_dose_mg)
print("Ibuprofen dose:", dose_mg, "mg")
# For 70 kg: 70 * 10 = 700, capped at 400 mg.
```

Solution 2 - dose in grams:

```
weight_kg = float(input("Enter patient weight in kg: "))
dose_mg = min(weight_kg * 15, 1000)
dose_g = dose_mg / 1000
print("Dose:", dose_mg, "mg =", dose_g, "g")
```

Solution 3 - include the patient's name:

```
name = input("Enter patient name: ")
weight_kg = float(input("Enter patient weight in kg: "))
dose_mg = min(weight_kg * 15, 1000)
print("Dose for", name + ":", dose_mg, "mg")
```

11. Knowledge check (answer in your head, then check)

Q1. What does print("Amoxicillin") show? A: Amoxicillin.

Q2. What does this show? x = 5, then print(x). A: 5 - print(x) with no quotes opens the box called x.

Q3. What error does print(price) give if no box called price exists? A: NameError.

Q4. A patient weighs 80 kg. What paracetamol dose does the program give, and how many

tablets? A: 80 x 15 = 1200, capped at 1000 mg, so 1000 mg = 2.0 tablets.

Q5. Why do we write float(input(...)) instead of just input(...)? A: input() always gives text; float() converts it to a number so maths works.

12. Summary

Lesson 1 taught four tools: variables (labelled boxes, = means store), print() (show something), input() (ask and wait), and float() (text to number). With two helpers, min() (pick the smaller, the safety ceiling) and division (tablets), those four tools build a real dose calculator. The program runs top to bottom, one line at a time.

Completion check - Lesson 1 is done when you can, without looking: write the dose calculator from scratch, run it for 60 kg (expect 900 mg, 1.8 tablets) and 70 kg (expect 1000 mg, 2.0 tablets), change it to ibuprofen (10 mg/kg, max 400 mg), and explain every line in your own words. That is exactly what was completed live on 26 Oct 2026.